21
Programming Assignment 3

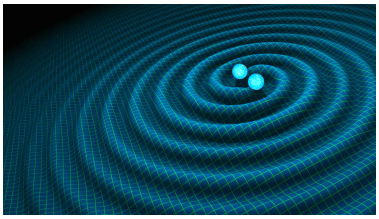


Figure 21.1: Artist's depiction of gravitational waves created by a merger of two neutron stars. Credits: US National Science Foundation.

In 2017, the Nobel Prize in physics was awarded to Rainer Weiss, Barry Barish and Kip Thorne for the discovery of gravitational waves. Only two years earlier, on September 14, 2015, 09:50:45 UTC, the two Laser Interferometer Gravitational-Wave Observatory (LIGO) instruments detected a gravitational wave for the first time in history. Gravitational waves were predicted by Einstein's general theory of relativity but had never been detected in situ before.

The first gravitational wave event detected by LIGO is thought to be created from a collision of two black holes, which sends out a localized chirp-like pulse in the space-time. LIGO utilizes two detectors, which are spaced 3000 km apart. These detectors measure *strain* ($\Delta L/L$) as a function of time. Here ΔL is the variation in length and L is the total length in which the variation is measured. In other words, strain is the normalized variation in length L of the interferometer line due to gravitational-wave. Figure 21.5 provides a high level overview of the measurement.

LIGO uses two geographically separated stations to measure gravitational waves: Hanford (H_1), and Livingston (L_1). The gravitational wave propagates at the speed of light $c \approx 3 \cdot 10^8$ m/s. If the same signal is detected at two different places with a time difference less than or equal to the speed of light propagation time between the sensors, then this provides more confidence that the event is in fact real, and not caused by for example local seismic activity. A third sensor would allow determining the direction of arrival based on time of arrival.



Figure 21.2: The Hanford and Livingston interferometers. Credits: LIGO.

The LIGO data is severely corrupted with instrumental noise. This noise is much larger in amplitude than the gravitational wave signal. However, this noise is very narrowband in nature, and it can be filtered out using relatively basic signal processing techniques without affecting the relatively broad band gravitational signal very much. Such filtering is routinely used by LIGO to improve the sensitivity of the instrument.

Verification of scientific results is an important part of science. UiT does not have the resources for building a giant interferometer, and this course really isn't about solving Einstein's field equations, so we won't be able to reproduce all the results. However, we can verify the signal processing part. In this programming assignment, your task is to develop signal processing software to verify that you can detect the gravitational wave signature in the LIGO measurements.

Instructions

We expect you to complete the listed signal processing tasks. The submission form is a written report that answers the questions given in each part of this assignment. The report should include the code with comments that indicate what each part of the program does. **The report should have at most ten pages**. You can return your report on Canvas by November 4th to receive feedback before the final portfolio submission on November 11th. The final portfolio submission including all three programming assignments will be on Wiseflow on November 11th.

You only need to know about signal processing concepts taught in FYS-2006. You do not need to know anything about gravitational waves or the LIGO instrument in order to complete the assignment. The lecture notes for the course contain several helpful signal processing examples. Take a look at lecture notes on *spectral analysis*, and *arbitrary frequency response filters*.

Instruction for reading the data

To begin, download the data files and a simple program that demonstrates reading these files from this location on the github repo for this course: `code/029_ligo`. The data files are:

```
H-H1_LOSC_4_V2-1126259446-32.hdf5
L-L1_LOSC_4_V2-1126259446-32.hdf5
```

These data files contain real measurements from LIGO starting at 2015-09-14T09:50:30 UTC.

After downloading the files, the next step is to read the strain signal from the data files. In Python, this can be done using the h5py module[1].

[1] Make sure you have the h5py module installed on your computer!

```
import h5py
h = h5py.File("file.hdf5","r")
data = h["strain/Strain"][()]
```

You will need to read two data vectors. One for the Livingston station (L_1) and one for the Hanford station (H_1). We will use the

symbol $x_H[n]$ of the Hanford signal and $x_L[n]$ for the Livingston signal.

1. Data

The sample rate of both of the signals is $f_s = 4096$ Hz. The samples in both signals are synchronized in time, i.e., sample n in signal $x_H[n]$ and $x_L[n]$ occur at the same time. Start by writing code to read the Hanford and Livingston signals from the data file.

a) How many samples are in each of the signals: $x_H[n]$ and $x_L[n]$?

b) How many seconds long is each signal $x_H[n]$ and $x_L[n]$ in the dataset?

2. Plotting the data

In order to see what the signals look like, you will need to plot the data.

a) Plot the signals $x_H[n]$ and $x_L[n]$, with time in seconds on the horizontal axis and strain on the vertical axis. Assume that time at the beginning of the signal array starts at 0 seconds. Label the axes of your plot. Use separate plots for $x_H[n]$ and $x_L[n]$ signals. Hint: you can use the `plt.plot(t,signal)` command found in Matplotlib. Use an array `t` to denote the seconds of each sample of the array `signal`.

3. Selecting a tapered window function

You will need to apply a discrete Fourier transform to analyze the spectral content of the signal. You will need to select a suitable tapered window function in order to obtain a good rejection of out of band signals. This will be crucial for detecting the real LIGO signal later. In this task, we'll use a synthetic narrowband signal to compare the performance of a discrete Fourier transform (DFT) based spectral analysis using a tapered window to spectral analysis done without a tapered window.

In order to calculate the magnitude spectrum of a signal $x[n]$ of length N, you need to evaluate a DFT on the signal:

$$\hat{x}[k] = \sum_{n=0}^{N-1} x[n] e^{-i\frac{2\pi}{N}kn}. \tag{21.1}$$

and the windowed signal:

$$\hat{x}_w[k] = \sum_{n=0}^{N-1} w[n]x[n] e^{-i\frac{2\pi}{N}kn}. \tag{21.2}$$

Hint: Use the `fft` function to evaluate the DFT. This function is available in Python as `numpy.fft.fft`.

a) Find Python functions in scipy that implement the Hann and Hamming windows. You can use tapering window functions available in the `scipy.signal` module.

b) In order to get an idea of how the window functions behave, apply it to a test signal consisting of two sinusoids. One with a weak amplitude and one with a strong amplitide.

$$x[n] = 10^{-5}\cos(2\pi f_1 n/f_s) + \cos(2\pi f_2 n/f_s) \tag{21.3}$$

Use a sample-rate of $f_s = 4096$ Hz. The frequencies of the sinusoids are $f_1 = 31.5$ Hz and $f_2 = 1234.56$ Hz, and the amplitudes 10^{-5} and 1 as shown in the above equation. Sample signal at sample indices $n \in [0, 1, 2, \cdots, 4095]$. Make a plot of the signal $x[n]$ and the windowed signal $w[n]x[n]$. Use a window of the same length as your signal $N = 4096$. Hint: You can use `numpy.arange(N)` to create a sequence of integers between 0 and $N-1$.

c) The FFT algorithm will evaluate $\hat{x}[k]$ at integer values of k between 0 and $N-1$. This corresponds to frequencies of $\hat{\omega}_k = 2\pi k/N$ in units of radians per sample. Provide a mathematical equation that converts these frequencies to hertz in the principal spectrum ($-f_s/2 < f_k < f_s/2$).

d) Estimate the power spectrum of the windowed signal $w[n]x[n]$ (with tapering) and the signal $x[n]$ (without tapering). Plot the power spectrum in decibel scale (power) for both. Use frequency in Hz on the horizontal axis and magnitude squared $10\log_{10}(|\hat{x}[k]|^2)$ (decibels) on the vertical axis. Plot both the positive and negative frequencies. Hint: You can use `numpy.fft.fftfreq` and `numpy.fft.fftshift` to determine what frequency (in hertz) each FFT bin k corresponds to, and to order the frequencies in ascending order.

e) Mark the locations of 31.5 and 1234.56 Hz in the plot.

f) Does the rectangular window (no tapering) allow you to detect the 31.5 Hz spectral line?

g) Compare the Hann and Hamming windows. Which one allows you to detect the weak signal better? Use the tapering window that works best in the next section.

4. Estimating the spectrum of the LIGO signal

a) Calculate the power spectrum of the LIGO signals $|\hat{x}_L[k]|^2$ and $|\hat{x}_H[k]|^2$. Here $\hat{x}_L[k]$ is the windowed DFT of the Livingston signal $x_L[n]$ and $\hat{x}_H[k]$ is the windowed DFT of the Hanford signal $x_H[n]$. The windowed DFT is obtained using

$$\hat{x}[k] = \sum_{n=0}^{N-1} w[n]x[n]e^{-i\frac{2\pi}{N}kn} \tag{21.4}$$

Perform the DFT over the whole dataset, i.e., N is the number of samples in the whole signal vector. Use the window function that you have chosen in the previous exercise, but make sure that use a window of length N, where N is the LIGO data vector length. Hint: Use FFT.

b) Plot the results with frequency in Hz in the horizontal axis and power using decibel scale on the vertical axis. Make separate plots for the Hanford and data Livingston. Plot only the positive frequencies.

c) On what frequencies are there strong spectral components in the Hanford and Livingston power spectra? Use Hz as the unit of frequency. Identify up to 12 frequency bands that contain narrowband interference. Label these regions on the plot of the power spectrum.

5. Whitening filter

In order to remove instrumental noise, you will next implement a whitening filter and apply it to the LIGO signal. A whitening filter is a filter that modifies the amplitudes of each spectral component in such a way that the magnitude spectrum of the filter output is constant-valued.

A whitening filter can be implemented as an FIR filter, which in frequency domain can be implemented as a multiplication:

$$\hat{y}[k] = \hat{h}[k]\hat{x}[k]. \tag{21.5}$$

Here $\hat{y}[k]$ is the DFT of the output of the filter, $\hat{h}[k]$ is the DFT of the whitening filter, and $\hat{x}[k]$ is the windowed DFT of the input signal $x[n]$.

The purpose of a whitening filter is to filter the signal in such a way that the magnitude of the output signal is unity: $|\hat{y}[k]| = 1$. This can be obtained using a filter of the form:

$$\hat{h}[k] = \frac{1}{|\hat{x}[k]|}. \tag{21.6}$$

a) Implement a whitening filter in frequency domain $\hat{h}[k]$ for the LIGO data. Implement a separate filter for the Hanford and Livingston signals. Use all the signal as input to the windowed DFT when calculating $\hat{x}[k]$. Do not filter the signal in smaller blocks!

b) Use an inverse discrete Fourier transform to transform the whitened signal $\hat{y}[k]$ into time-domain. Do this for both Livingston and Hanford signals separately (obtaining $y_L[n]$ and $y_H[n]$).

c) Plot the whitened signals $y_H[n]$ and $y_L[n]$ for both Hanford and Livingston. Use x-axis for time in seconds ($t \in [0, t_{\max}]$) and y-axis for whitened strain $y[n]$. The gravitational wave signal is in the middle of the signal, between 16.2 and 16.5 seconds. If you've done everything correctly, you should see the gravitational wave signal. It looks like a chirp (see plot in Figure 21.3). However, because we are not done yet, the signal will look a lot noisier than the Figure 21.3.

6. Low-pass filtering

The gravitational wave signal is at frequencies below 300 Hz. Design a simple running mean averaging low-pass filter

$$y[n] = \frac{1}{L} \sum_{k=0}^{L-1} x[n-k] \tag{21.7}$$

that will attenuate spectral components with frequencies above 300 Hz.

a) Find an integer value of L such that the filter will reduce the power of frequency components at $f = 300$ Hz by approximately -6 dB compared to the filter output for a $f = 0$ Hz signal.

b) Plot the power spectral response of the filter in dB scale ($10 \log_{10} |\mathcal{H}(\hat{\omega})|^2$), where $\mathcal{H}(\hat{\omega})$ is the discrete-time Fourier transform of the FIR filter coefficients of the averaging filter. Use frequency on the horizontal axis in Hz. Label the -6 dB point in frequency and in power spectral response, verifying that the -6 dB point is close to 300 Hz.

c) What is the time delay τ to the signal introduced by the filter, in seconds?

d) Apply the running mean average low-pass filter on the whitened Hanford and Livingston signals ($y_H[n]$ and $y_L[n]$).

e) Undo the effects of the filter time-delay by shifting the signal in time. You can do this by adjusting the time variable $t' = t - \tau$, instead of filtering the signal.

f) Plot the low pass filtered whitened Hanford and Livingston signals. You should see the gravitational wave signal more clearly now. Use the horizontal axis for time and the vertical axis for the signal amplitude. Plot only the time interval between 16.1 and 16.6 seconds where the gravitational wave signal is located. Compare your plot with Figure 21.3. Your plot should be similar, but not necessarily exactly the same, as your filter is not exactly the same as the one used for Figure 21.3.

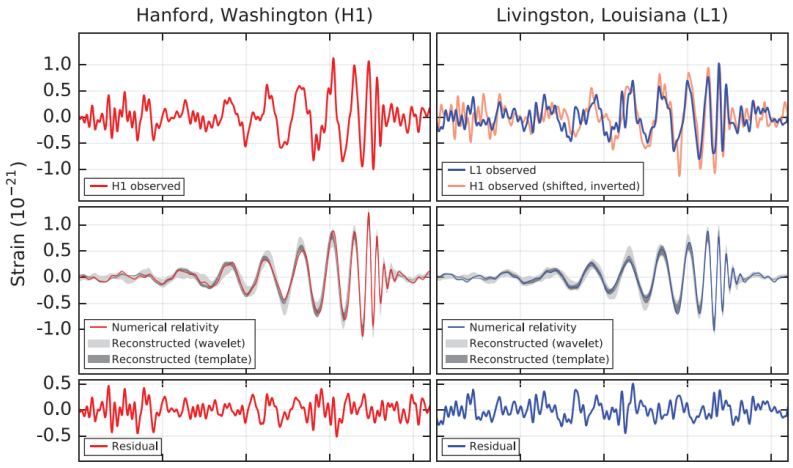


Figure 21.3: The figure is from: B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration) Phys. Rev. Lett. 116, 061102 – Published 11 February 2016

7. *Time delay*

Gravitational waves are expected to propagate at the speed of light. The Hanford and Livingston detectors are separated by about 3000 km. A gravitational wave will propagate this distance in about 10 ms.

The gravitational wave angle of arrival is not known, but the relative time delay between the signal detected at Hanford and Livingston is expected to be $-10 < \tau < 10$ ms, if it is moving at the speed of light in vacuum.

a) Determine the time separation between the two signals *by plotting the magnitudes* of the filtered gravitational wave signals ($|y_L[n]|$ and $|y_H[n + n_0]|$) with different delays n_0 on the same plot. Try different values of n_0 until the signals are approximately aligned in time. The magnitude of the signal is used to avoid phase-time

ambiguities. Hint: Use magnitude, not amplitude. Magnitude can be obtained using `numpy.abs`.

b) What value do you obtain for the sample delay n_0?

c) What time delay τ in seconds does the sample delay n_0 correspond to?

d) Is the time delay τ in agreement with gravitational-wave propagation speed (i.e., that $-10 < \tau < 10$ ms)?

8. Dynamic spectrum

Study the time-frequency behavior of the gravitational wave signal.

a) Calculate the dynamic spectrum (spectrogram) of the low-pass filtered whitened signal $|\hat{x}[t,k]|$, where t is time and k is frequency.

b) Plot the dynamic power spectrum $|\hat{x}[t,k]|^2$ in dB scale. Perform this in such a way that you can see the time-frequency response behavior of the gravitational wave signal clearly. Plot your result at between $t = 15.5$ s and $t = 17$ s. Hint: You will need to use zero-padding, a tapered window, and overlapping windows. An example of the time-frequency domain behavior of the gravitational wave signal is shown in Figure 21.4.

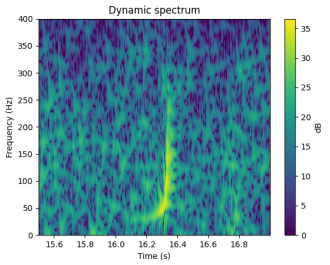


Figure 21.4: A dynamic spectrum plot of the gravitational wave signal.

9. Extra task

For earning an extra point, improve any part of the signal processing in a way that you see fit. Document your improvements. You can, e.g., try to create a filter that removes only the strong frequency components from the signal, or you can use a better low-pass filter.

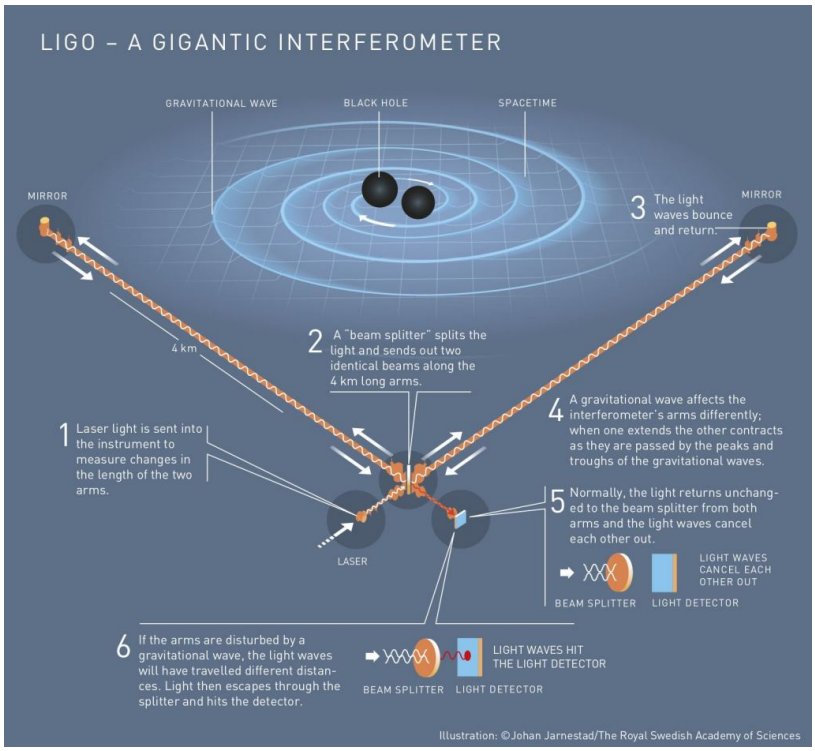


Figure 21.5: The gravitational wave measurement using a Michelson-Morley interferometer. Credits: Johan Jarnestad, The Royal Swedish Academy of Sciences.